GLANSIS ORGANISM RISK ASSESSMENT

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SPECIES INFORMATION

Scientific Name: *Procambarus virginalis*

Common Name: Marbled crayfish, Marmokrebs

SECTION A: INTRODUCTION POTENTIAL RESULTS

Dispersal: Moderate
Hitchhiking/Fouling: High
Unauthorized intentional release: High
Stocking/Planting/Escape from recreational culture: Low
Escape from commercial culture: Unknown
Shipping: Low

***Procambarus virginalis* has a high probability of introduction to the Great Lakes (Confidence level: Moderate).**

Comments:

The most likely pathways of introduction for *Procambarus virginalis* include unauthorized intentional release, hitchhiking/fouling, and dispersal. *P. virginalis* is a popular pet species that has been in the North American pet trade since 2004, it can be purchased online and in person from hobbyists (Chucholl 2010; Faulkes 2010). Regulations exist in the region that restrict the possession and sale of marbled crayfish but the recent conviction of a woman in Ohio for selling this species suggest introductions from the pet trade remain a concern (Milman 2022). The persistence of a *P. virginalis* population in Burlington, Ontario (Hamr 2023) increases the risk of introduction via dispersal and hitchhiking. *P. virginalis* are able to travel long distances over land (Guo et al. 2019; Vogt et al. 2019) and there are documented instances of active dispersal in Europe (Chucholl et al. 2012); this suggests a potential for further dispersal from the stormwater ponds in Ontario. Additionally, given the close proximity of the Ontario population, the potential exists for waterbird-mediated dispersal, bait bucket transfer, and other forms of hitchhiking introductions (Jones et al. 2009; Anastácio et al. 2014; Vogt 2021).

POTENTIAL INTRODUCTION VIA DISPERSAL

A1) Does this species occur near waters (natural or artificial) connected to the Great Lakes basin* (e.g., streams, ponds, canals, or wetlands)? (*Great Lakes basin = below the ordinary high water mark, including connecting channels, wetlands, and waters ordinarily attached to the Lakes)

Yes, this species occurs near waters connected to the Great Lakes basin and is mobile or able to be transported by wind or water.	100 √
No, this species does not occur near waters connected to the Great Lakes basin and/or is not mobile or able to be transported by wind or water.	0
Unknown	U

- *There is one research grade iNaturalist observation of marbled crayfish Procambarus virginalis in New York from 2023, but this specimen was observed in a brook that is not near waters connected to the Great Lakes basin (Dinadublon 2023). Additionally,* P. virginalis *are readily available for purchase as lab or aquarium specimens and as fishing bait in the US (Chucholl 2010).*
- *In October 2021, there were reports of suspected* Procambarus virginalis *from a park in Burlington, Ontario. All reports occurred near the stormwater ponds in the park, further investigation confirmed the identity as marbled crayfish (Hamr 2023). Eradication efforts took place in the winter of 2022-2023, but continued reports suggest they were unsuccessful (Chandler 2023). The stormwater ponds in the park are hydrologically isolated and the population appears to be confined to the park at the moment (Hamr 2023).*

A2) What is the proximity of this species to the Great Lakes basin?

This species occurs in waters within 20 kilometers of the Great Lakes basin, and no barrier (e.g., electric barrier, dam) to dispersal is present.	Score x 1
This species occurs in waters within 20 kilometers of the Great Lakes basin, but dispersal to the basin is blocked; or, this species occurs in waters within 100 kilometers of the Great Lakes basin, and no barrier to dispersal is present.	Score x 0.75 √
This species occurs in waters within 100 kilometers of the Great Lakes basin, but dispersal to the basin is blocked.	Score x 0.5
This species occurs in waters >100 kilometers from the Great Lakes basin.	Score x 0.25
Unknown	U

- *The* Procambarus virginalis *observation in New York is approximately 350 km from Lake Ontario while the population in Burlington, Ontario is only 6 km away from Lake Ontario. In both instances the inhabited water bodies are isolated from waters connected to the Great Lakes basin.* P. virginalis *are able to travel long distances over land (Guo et al. 2019; Vogt et al. 2019) and there are documented instances of active dispersal in Europe (Chucholl et al. 2012); this suggests a potential for further dispersal from the stormwater ponds in Ontario.*

POTENTIAL INTRODUCTION VIA HITCHHIKING/FOULING

A3) Is this species likely to attach to or be otherwise transported by, or along with, recreational gear, boats, trailers, fauna (e.g., waterfowl, fish, insects), flora (e.g., aquatic plants), or other objects (e.g., packing materials), including as parasites or pathogens, entering the Great Lakes basin?

Yes, this species is known to be able to adhere to certain surfaces or to be transported by other organisms entering the Great Lakes basin.	100 √
No, this species is not known to be able to adhere to certain surfaces or to be transported by other organisms entering the Great Lakes basin.	0
Unknown	U

- *Waterbird-mediated dispersal is possible and has been documented in other species of invasive crayfish (Anastácio et al. 2014), but has not been observed with marbled crayfish.*
- *There is speculation that* Procambarus virginalis *was introduced to Madagascar by foreign contractors working on a road construction project (Jones et al. 2009) but the specifics of this introduction remain unknown.*
- *Crayfish are popular bait for recreational fishing and marbled crayfish hitchhiking via bait buckets remains a possibility, its spread throughout Madagascar is partially attributed to fishing activities (Vogt 2021).*

A4) What is the proximity of this species to the Great Lakes basin?

This species occurs in waters within 20 km of the Great Lakes basin.	Score x 1 √
This species occurs in waters within 100 km of the Great Lakes basin.	Score x 0.5
This species occurs in waters >100 km from the Great Lakes basin.	Score x 0.1
Unknown	U

- *The* Procambarus virginalis *observation in New York is approximately 350 km from Lake Ontario while the population in Burlington, Ontario is only 6 km away from Lake Ontario.*

POTENTIAL INTRODUCTION VIA UNAUTHORIZED INTENTIONAL RELEASE

A5) Is this species sold at aquarium/pet/garden stores ("brick & mortar" or online), catalogs, biological supply companies, or live markets (e.g., purchased for human consumption, bait, ornamental, ethical, educational, or cultural reasons) and as a result may be released into the Great Lakes basin?

Yes, this species is available for purchase.	100 √
No, this species is rarely/never sold.	0
Unknown	U

- *Aquarist and bait stores frequently sell marbled crayfish at very low prices and ship them around the country, as they make for popular pets. Crayfish enthusiasts may also send specimens to each other, facilitating their spread (Chucholl 2010).*
- Procambarus virginalis *has been in the North American pet trade since 2004, they are available through online sources and acquisition through face-to-face personal contact is not uncommon (Faulkes 2010).*
- P. virginalis *is a popular and inexpensive aquarium pet, they are available for purchase from hobbyists in the United States for about $5-25 per specimen (The Shrimp Farm 2023).*
- *As a popular organism for research,* Procambarus virginalis *can be used in university or government research labs. Potential introduction via biological supply trade or lab escape as a result.*

A6) How easily is this species obtained within the Great Lakes region (states/provinces)?

This species is widely popular, frequently sold, and/or easily obtained within the Great Lakes region.	Score x 1
This species is widely popular, and although trade, sale, and/or possession of this species is prohibited, it is frequently sold on the black market within the Great Lakes region.	Score x 0.5 √
This species is not very popular or is not easily obtained within the Great Lakes region.	Score x 0.1
Unknown	U

- *Marbled crayfish are readily available for purchase online in the United States, and will ship within the Great Lakes region (Faulkes 2010).*
- *In 2022, an Ohio woman was charged under the Lacey Act for selling marbled crayfish online. Media reports indicate she sold marbled crayfish to people across 36 states (Milman 2022).*

POTENTIAL INTRODUCTION VIA STOCKING/PLANTING OR ESCAPE FROM RECREATIONAL CULTURE

A7) Is this species being stocked/planted to natural waters or outdoor water gardens around the Great Lakes region?

Yes, this species is being stocked/planted and/or has ornamental, cultural, medicinal, environmental (e.g., biocontrol, erosion control), scientific, or recreational value in the Great Lakes region.	100 √
No, this species cannot be stocked/planted or there is not enough interest to do so in the Great Lakes region.	0
Unknown	U

- *In Europe, marbled crayfish are a popular food source for ornamental turtles (Chucholl, 2010). Since ornamental turtles may be kept in open ponds, this use of marbled crayfish may facilitate accidental introductions, but no records exist of this phenomenon in the United States.*
- *While there is no documentation regarding stocking of* Procambarus virginalis *in the Great Lakes region, this species has ornamental, scientific, and recreational value. Marbled crayfish are one of many crayfish species that are kept in regions where ornamental fishkeeping is a popular hobby, including the USA (Blaha et al. 2022). Due to its ease of culture, handling tolerance, fertility, adaptability, and genetic uniformity, this species has scientific value as a model organism for scientific research (Hossain et al. 2018).* P. virginalis *has recreational value as a popular pet species and its use as bait for fishing (Vogt 2021; Lipták et al. 2023).*

A8) What is the nature and proximity of this activity to the Great Lakes basin?

This activity is authorized and/or is occurring directly in the Great Lakes.	Score x 1
This activity is occurring in Great Lakes tributaries or connecting waters, or within 20 km of the Great Lakes basin.	Score x 0.75
This activity is <u>likely</u> to occur within 20 km of the Great Lakes basin because of its popularity/value and there are no widespread regulations against stocking/planting.	Score x 0.5
This activity is occurring in waters >20 km from the Great Lakes basin, or despite federal or state regulations in more than half the basin (> 5 states/provinces), this activity <u>may</u> occur within 20 km of the basin because of the species' popularity/value.	Score x 0.25 √
Unknown	U

- *At the time of this review, no records of deliberate* Procambarus virginalis *stocking exist in the Great Lakes region. However, given this crayfish's well-documented ornamental use and popularity as a pet (Chucholl 2010; Faulkes 2010; Blaha et al. 2022; Lipták et al. 2023), this activity may occur within 20 km of the Great Lakes basin. In 2022, for example, an Ohio woman was charged under the Lacey Act for selling marbled crayfish online. Media reports indicate she sold marbled crayfish to people across 36 states (Milman 2022), suggesting a demand for this species nationwide.*

POTENTIAL INTRODUCTION VIA ESCAPE FROM COMMERCIAL CULTURE

A9) Is this species known to be commercially cultured in or transported through the Great Lakes region?

Yes, this species is being commercially cultured in or transported through the Great Lakes region.	100

No, this species is not commercially cultured in or transported through the Great Lakes region.	0
Unknown	U √

- *Commercial culture for aquaria does exist in the US, but typically in the southern states: the frequency of transportation of marbled crayfish through the Great Lakes is unknown.*

A10) What is the nature and proximity of this activity to the Great Lakes basin?

This activity is unregulated or minimally regulated and is occurring directly in the Great Lakes.	Score x 1
This activity is unregulated or minimally regulated and is occurring in Great Lakes tributaries or connecting waters, or within 20 km of the Great Lakes basin.	Score x 0.75
This activity is strictly regulated but occurs directly in the Great Lakes, and/or this activity involves transport of live organisms on/across the Great Lakes.	Score x 0.5
This activity is strictly regulated but occurs in Great Lakes tributaries, connecting waters, or within 20 km of the Great Lakes basin, and/or this activity involves transport of live organisms within 20 km of the Great Lakes basin.	Score x 0.25
This activity occurs >20 km from the Great Lakes basin and typically does not involve transport of live organisms closer to the basin.	Score x 0.1
Unknown	U √

- *The frequency of transportation of marbled crayfish through the Great Lakes is unknown.*

POTENTIAL INTRODUCTION VIA SHIPPING

A11) Is this species capable of surviving adverse environments (i.e. extreme temperatures, absence of light, low oxygen levels) and partial-to-complete ballast water exchange (BWE) (e.g., is euryhaline, buries in sediment, produces resistant resting stages, has other attributes or behaviors facilitating survival under these conditions)?

Yes, this species is able to survive in ballast tank environments for weeks at a time and may be suspended in ballast water.	100
Yes, this species is able to survive in ballast tank environments for weeks at a time and is able to survive BWE by burial in ballast sediment.	80
Yes, this species is able to survive in ballast tank environments for weeks at a time and may be suspended in ballast water, but this species is not able to survive BWE.	60 √
No, but this species is capable of fouling transoceanic ship structures (e.g., hull, chains, chain locker) while in its active or resting stage.	40
No, this species is not able to survive adverse environments, does not foul transoceanic ship structures, or is unlikely to be taken up with ballast.	0
Unknown	U

- *Results of a lab experiment examining* Procambarus virginalis *ontogeny show that this species can withstand temperatures below 8 °C and above 30 °C, individuals survived at these temperatures for longer than 40 days (Seitz et al. 2005). The warmest reported habitat for* P. virginalis *is Lake Hevíz, Hungary with water temperatures of 38 °C in summer while the coldest habitat is the Marstaån River, Sweden where winter water temperatures are around 0 °C (Vogt et al. 2019).*
- *Salinity tolerance of* P. virginalis *was found to be lower than other crayfish species, but it can survive for more than 80 days at 18 ppt (Veselý et al. 2017).*
- *Marbled crayfish withstands desiccation and drought through vertical burrowing (Kouba et al. 2016) and is likely capable of burrowing in sediment.*

- P. virginalis *can occur in both acidic and alkaline waters and requires an oxygen concentration 2.9 mg/l or greater (Wiśniewski et al. 2020).*
- *While marbled crayfish can survive adverse environments, there are no reports of it fouling transoceanic ship structures. This freshwater species is highly unlikely to be able to survive ballast water exchange.*

A12) Does this species occur in waters from which shipping traffic to the Great Lakes originates?

Yes, and this species has been observed in the ballast of or fouling ships entering the Great Lakes.	Score x 1
Yes, and this species has been observed in ports that have direct trade connections with the Great Lakes (e.g., Baltic Sea).	Score x 0. 5
Yes, but this species has neither been observed in ballast/fouling ships entering the Great Lakes nor in ports in direct trade with the Great Lakes.	Score x 0.1√
No, this species does not occur in waters from which shipping traffic to the Great Lakes originates.	Score x 0
Unknown	U

- Procambarus virginalis *global distribution includes Italy, Belgium, and Sweden (Scheers et al. 2021; Chucholl 2022), there are vessel routes to the Great Lakes from all of these countries (EPA 2015). However, there are no reports of this species fouling transoceanic vessels and it inhabits streams and sloughs, not the deeper water environments necessary for shipping (Chucholl 2010).*

Potential Vector Ranking and Points

Vector	**Raw Points Scored**	**Proximity Multiplier**	**Total Points Scored**	**Probability of Introduction**
Dispersal: Natural dispersal through waterbody connections or wind	**100**	x 0.75	**75**	Moderate
Hitchhiking/Fouling: Transport via recreational gear, boats, trailers, mobile fauna, stocked/planted organisms, packing materials, host organisms, etc.	**100**	x 1	**100**	High
Release: Unauthorized intentional release of organisms in trade (e.g., aquaria, water gardens, live food)	**100**	x 1	**100**	High
Stocking/Planting/Escape from recreational culture: Intentional authorized or unauthorized introduction to natural waters in the Great Lakes *OR* Accidental introduction to Great Lakes by escape from recreational culture (e.g., water gardens)	**100**	x 0.25	**25**	Low
Escape from commercial culture: Accidental introduction to Great Lakes by escape from commercial culture (e.g., aquaculture)	**U**	x U	**U**	Unknown
Trans-oceanic shipping: Ballast (BOB) or no-ballast-on-board (NOBOB) water	**60**	x 0.1	**6**	Low

exchange/discharge, sediment discharge, hull fouling					
Total Unknowns (U)	2	Confidence Level		Moderate	

Qualitative Statements for GLANSIS Fact Sheet:

Procambarus virginalis has a high probability of introduction to the Great Lakes (Confidence level: Moderate).

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Potential pathway(s) of introduction: Unauthorized intentional release, Dispersal, Hitchhiking/Fouling

Scoring	
Points (per vector)	**Probability for Introduction**
80-100	High
40-79	Moderate
1-39	Low
0	Unlikely
# of Unknowns (overall)	**Confidence Level**
0	High
1-2	Moderate
3-5	Low
>5	Very low

SECTION B: RISK OF ESTABLISHMENT RESULTS

INVASIVE BIOLOGICAL/ECOLOGICAL ATTRIBUTES

B1) How would the physiological tolerance of this species (survival in varying temperature, salinity, oxygen, and nutrient levels) be described?

This species has broad physiological tolerance. It has been reported to survive in wide ranges of temperature (0°C-30°C), salinity (0-16 parts per thousand), oxygen (0-saturated), AND nutrient (oligotrophic-eutrophic) levels.	9 √
This species has somewhat broad physiological tolerance. It has been reported to survive in a wide range of temperature, salinity, oxygen, OR nutrient levels. Tolerance to other factors is narrower, unknown, or unreported.	6
This species has narrow physiological tolerance. It has been reported to survive in limited ranges of temperature, salinity, oxygen, and nutrient levels.	3
Unknown	U
	6

- *Results of a lab experiment examining Procambarus virginalis ontogeny show that this species can withstand temperatures below 8 °C and above 30 °C, individuals survived at these temperatures for longer than 40 days (Seitz et al. 2005). The warmest reported habitat for P. virginalis is Lake Hevíz, Hungary with water temperatures of 38 °C in summer while the coldest habitat is the Marstaån River, Sweden where winter water temperatures are around 0 °C (Vogt et al. 2019).*
- *Found in both lentic and lotic conditions,temporary wetlands, and oligotrophic to eutrophic lakes (Kawai et al., 2009; Vogt et al. 2019).*
- *Aquarium hobbyists indicate that this species can tolerate variance in pH and oxygen, and can even tolerate water "straight from the tap" (Robbins, 2018).*
- *P. virginalis can occur in both acidic and alkaline waters and tolerates an oxygen concentration 2.9 mg/l or greater (Wiśniewski et al. 2020).*
- *Salinity tolerance of P. virginalis was found to be lower than other crayfish species, but it can survive for more than 80 days at 18 ppt (Veselý et al. 2017). Additionally, this species can survive in brackish waters for months, but does not reproduce in these conditions (Vogt 2021).*
- *Marbled crayfish withstands desiccation and drought through vertical burrowing (Kouba et al. 2016).*

B2) How likely is it that any life stage of this species can overwinter in the Great Lakes (survive extremely low levels of oxygen, light, and temperature)?

Likely (This species is able to tolerate temperatures under 5°C and oxygen levels ≤0.5 mg/L)	9 √
Somewhat likely (This species is able to tolerate some of these conditions OR has adapted behaviorally to avoid them)	6
Somewhat unlikely (This species is able to tolerate conditions close to those specified, but it is not known as an overwintering species)	3
Unlikely	0
Unknown	U
	9

- Procambarus virginalis *can survive ice cover (Chucholl and Pfeiffer 2010) and its current distribution in Poland, Sweden, and Ontario suggest a similar overwintering potential in the Great Lakes.* P. virginalis *is able to survive over winter in Polish reservoirs with an ice cover (Wiśniewski et al. 2020). Individuals were*

observed crawling on the substrate in the River Marstaån, Sweden where water temperatures were 0°C (Vogt et al. 2019). In Burlington, Ontario, a population of P. virginalis *has persisted over two winters (2021-2022 and 2022-2023) despite the dewatering of ponds in an attempt to freeze them out (Hamr 2023).*

B3) If this species is a heterotroph, how would the flexibility of its diet be described?

This species is a dietary generalist with a broad, assorted, AND flexible diet.	9 √
This species is moderately a dietary generalist with a broad, assorted, OR flexible diet.	6
This species is a dietary specialist with a limited and inflexible diet.	3
This species is an autotroph.	0
Unknown	U
	9

- *Like many crayfish species,* Procambarus virginalis *is a polytrophic omnivore, and feeds on detritus, algae, plants, and invertebrates (Souty-Grosset et al., 2006). Stable isotope analysis suggests that detritus was the most important component of* P. virginalis *diet, but zoobenthos, algae, and macrophytes were also consumed (Lipták et al. 2019). The marbled crayfish has an adaptive, generalist diet and is capable of occupying a wide range of trophic positions (Linzmaier et al. 2020).*

B4) How likely is this species to outcompete species in the Great Lakes for available resources?

Likely (This species is known to have superior competitive abilities and has a history of outcompeting other species, AND/OR available literature predicts it might outcompete native species in the Great Lakes)	9 √
Somewhat likely (This species is known to have superior competitive abilities, but there are few reported cases of this species outcompeting another and no predictions regarding species in the Great Lakes)	6
Somewhat unlikely (This species has average competitive abilities, and there are no reported cases of this species outcompeting another and no predictions regarding species in the Great Lakes)	3
Unlikely (This species is known as a poor competitor that thrives only in environments with low biodiversity, AND/OR available literature predicts it might be outcompeted by a species in the Great Lakes)	0
Unknown	U
	9

- *Marbled crayfish may pose a threat to indigenous crayfish due to competition for food and rapid reproduction (Jones et al., 2009; Chucholl and Pfeiffer, 2010).*
- *A lab experiment examining agonistic interactions between marbled crayfish* (Procambarus virginalis) *and red swamp crayfish* (Procambarus clarkii) *suggests that marbled crayfish has a competitive advantage. Marbled crayfish won significantly more fights and more frequently established dominance in both premature and mature experimental trials (Hossain et al. 2019).*
- *Marbled crayfish has shown competitive dominance over calico crayfish (*Faxonius immunis*) in experimental lab studies. When no shelter was available, marbled crayfish won significantly more fights. With available shelter, marbled crayfish dominance was 100% for female and 54.5% for male calico crayfish (Hossain et al. 2020).*

B5) How would the fecundity of this species be described relative to other species in the same taxonomic Class?

Very high	9 √
High	6
Moderate	3
Low	0
Unknown	U
	9

- Procambarus virginalis *is parthenogenetic and single individuals can reproduce without a mate, making them exceptionally fecund (Kawai et al., 2009).*
- P. virginalis *can produce up to 40% more pleopodal eggs than the congeneric red swamp crayfish* P. clarkii *(Sheppard et al. 2023).*

B6) How likely are this species' reproductive strategy and habits to aid establishment in new environments, particularly the Great Lakes (e.g., parthenogenesis/self-crossing, self fertility, vegetative fragmentation)?

Likely (The reproductive strategy or habits of this species are known to aid establishment in new environments, AND available literature predicts establishment in the Great Lakes based on these attributes)	9
Somewhat likely (The reproductive strategy or habits of this species are known to aid establishment in new environments, but there is no literature available regarding establishment in the Great Lakes based on these attributes)	6 √
Somewhat unlikely (The reproductive strategy or habits of this species could potentially aid establishment in new environments, but there is no literature available regarding establishment in the Great Lakes based on these attributes)	3
Unlikely (The reproductive strategy or habits of this species are not known to aid establishment in new environments)	0
Unknown	U
	6

- Procambarus virginalis *reproduces parthenogenetically, first reproduction typically occurs at an age of 5-6 months and a total length of 4 cm (Vogt 2008). Parthenogenetic reproduction means a single individual can establish a new population without the need for a mate (Chucholl 2010).*
- P. virginalis *has a relatively fast growth rate and frequent reproduction, both of which contribute to its ability to establish in new environments (Kouba et al. 2021).*

ENVIRONMENTAL COMPATIBILITY

B7) How similar are the climatic conditions (e.g., air temperature, precipitation, seasonality) in the native and introduced ranges of this species to those in the Great Lakes region?

Very similar (The climatic conditions are practically identical to those of the Great Lakes region)	9 √
Similar (Many of the climatic conditions are similar to those of the Great Lakes region)	6

Somewhat similar (Few of the climatic conditions are similar to those of the Great Lakes region)	3
Not similar	0
Unknown	U
	9

- *The marbled crayfish originates from P. fallax, which is native to Florida and Georgia (Dorn and Violin, 2009). Genetic analysis suggests that* P. virginalis *originated in northern Union County, Florida and both parental haplotypes were inherited from the Evergraldes subpopulation of* P. fallax *(Vogt et al. 2018; Gutekunst et al. 2021). No indigenous marbled crayfish has been reported, however, which makes it difficult to assess "native" environmental conditions to compare them to the Great Lakes region, but inferences can be made from its introduced range.*
- *The introduced range of* P. virginalis *in Europe includes Belgium, Netherlands, Germany, Hungary, Sweden, Poland, Ukraine, Slovakia, and Romania (Parvulescu et al. 2017; Lipták et al. 2017; Wiśniewski et al. 2020; Son et al. 2020; Weiperth et al. 2020; Scheers et al. 2021; Chucholl 2022). These areas have similar climatic conditions as the Great Lakes region in terms of temperature, seasonality, and precipitation. Additionally, the introduced population in Burlington, Ontario (Chandler 2023) has climatic conditions that are nearly identical to the Great Lakes.*
- *According to a climate suitability analysis performed by the Fish and Wildlife Service,* Procambarus virginalis *has a medium to high climate match across a majority of the US and the Great Lakes region is noted as having a high match. A climate match of 0.103 and higher is considered a high match, a score greater than 0.005 and less than 0.103 is considered a medium match, while a score less than or equal to 0.005 is categorized as a low match. Marbled crayfish has a match of 0.663 (USFWS 2023).*

B8) How similar are other abiotic factors that are relevant to the establishment success of this species (e.g., pollution, water temperature, salinity, pH, nutrient levels, currents) in the native and introduced ranges to those in the Great Lakes?

Very similar (These factors are practically identical to those of the Great Lakes region)	9 √
Similar (Many of these factors are similar to those of the Great Lakes region)	6
Somewhat similar (Few of these factors are similar to those of the Great Lakes region)	3
Not similar	0
Unknown	U
	9

- *No indigenous marbled crayfish has been reported, which makes it difficult to assess "native" abiotic conditions to compare them to the Great Lakes region, but inferences can be made from its introduced ranges.*
- Procambarus virginalis*, in its introduced ranges, has demonstrated high plasticity towards abiotic factors and tolerates variation in ecological parameters (Andriantsoa et al. 2019).*
- *In Madagascar, populations of* P. virginalis *have been observed at a wide range of altitudes (3-1491 m) and in areas with high concentrations of dissolved solids associated with pollution (Andriantsoa et al. 2019).*
- *In Hungary,* P. virginalis *has been observed in a thermal lake with water temperatures of 38 ℃ in summer while a population in Sweden was observed in a river with a water temperature around 0 ℃ in winter (Vogt et al. 2019). In addition,* P. virginalis *survives overwinter in Polish reservoirs with an ice cover and water temperatures of 4 ℃ (Wiśniewski et al. 2020).*
- *Marbled crayfish can tolerate brackish water conditions (0.1-1% salinity) for many months (Vogt 2021).*
- *Wiśniewski et al. (2020) report that* P. virginalis *has a wide range of pH tolerance and can occur in both acidic and alkaline waters.*

- Procambarus virginalis *has been observed in rivers, slow-moving streams, and lakes in its introduced range throughout Europe. This species can inhabit oligotrophic to eutrophic nutrient regimes and a range of current conditions (Vogt et al. 2019).*
- *Notably, the persistence of a* P. virginalis *population in Burlington, Ontario (Hamr 2023) suggests they can tolerate the abiotic conditions present in the Great Lakes region.*

B9) How abundant are habitats suitable for the survival, development, and reproduction of this species in the Great Lakes area (e.g., those with adequate depth, substrate, light, temperature, oxygen)?

Abundant (Suitable habitats can be easily found and readily available)	9
Somewhat abundant (Suitable habitats can be easily found but are in high demand by species already present)	6 √
Somewhat scarce (Suitable habitats can be found occasionally)	3
Scarce (Suitable habitats are rarely found)	0
Unknown	U
	6

- *Based on temperature tolerances in introduced ranges, flexibility in substrate and current selection, as well as moderate oxygen requirements (Vogt et al. 2019; Wiśniewski et al. 2020), suitable habitat for* Procambarus virginalis *is expected to be somewhat abundant in the Great Lakes area.*

B10) How likely is this species to adapt to or to benefit from the predicted effects of climate change on the Great Lakes freshwater ecosystems (e.g., warmer water temperatures, shorter duration of ice cover, altered streamflow patterns, increased salinization)?

Likely (Most of the effects described above make the Great Lakes a better environment for establishment and spread of this species, OR this species could easily adapt to these changes due to its wide environmental tolerances)	9
Somewhat likely (Several of the effects described above could make the Great Lakes a better environment for establishment and spread of this species)	6 √
Somewhat unlikely (Few of the effects described above would make the Great Lakes a better environment for establishment and spread of this species)	3
Unlikely (Most of the effects described above would have no effect on establishment and spread of this species or would make the environment of the Great Lakes unsuitable)	0
Unknown	U
	6

- *Warming of the Great Lakes and shorter duration of ice cover would be highly beneficial for* Procambarus virginalis*, increased temperatures has been shown to alter its reproductive cycle by increasing frequency and duration of gravid states (Sheppard et al. 2023)..*
- *Dynamic energy budget models predict that climate change will increase the competitive advantage of* P. virginalis *(Marn et al. 2022).*

B11) How likely is this species to find an appropriate food source (prey or vegetation in the case of predators and herbivores, or sufficient light or nutrients in the case of autotrophs)?

Likely (All possible nutritive food items—including species in the Great Lakes that may be considered potential food items—are highly abundant and/or easily found)	9 √

Somewhat likely (Some nutritive food items—including species in the Great Lakes that may be considered potential food items—are abundant and/or search time is low to moderate)	6
Somewhat unlikely (Few nutritive food items—including species in the Great Lakes that may be considered potential food items—are abundant and/or search time is moderate to high)	3
Unlikely (All possible nutritive food items—including species in the Great Lakes that may be considered potential food items—are relatively scarce and/or search time is high)	0
Unknown	U
	9

- *Like many crayfish species,* Procambarus virginalis *is a polytrophic omnivore, and feeds on detritus, algae, plants, and invertebrates (Souty-Grosset et al., 2006). Stable isotope analysis suggests that detritus was the most important component of* P. virginalis *diet, but zoobenthos, algae, and macrophytes were also consumed (Lipták et al. 2019). The marbled crayfish has an adaptive, generalist diet and is capable of occupying a wide range of trophic positions (Linzmaier et al. 2020).*

B12) Does this species require another species for critical stages in its life cycle such as growth (e.g., root symbionts), reproduction (e.g., pollinators, egg incubators), spread (e.g., seed dispersers), or transmission (e.g., vectors)?

Yes, and the critical species (or one that may provide a similar function) is common in the Great Lakes and can be easily found in environments suitable for the species being assessed; OR, No, there is no critical species required by the species being assessed	9 √
Yes, and the critical species (or one that may provide a similar function) is moderately abundant and relatively easily found in particular parts of the Great Lakes	6
Yes, and the critical species (or one that may provide a similar function) is relatively rare in the Great Lakes AND/OR can only be found occasionally in environments suitable for the species being assessed	3
Yes, and the critical species (or one that may provide a similar function) is not present in the Great Lakes but is likely to be introduced	0
Yes, but the critical species (or one that may provide a similar function) is not present in the Great Lakes and is not likely to be introduced	-80% total points (at end)
Unknown	U
	9

- *No, this species does not require the presence of any other species to grow, reproduce, or spread.*

B13) How likely is the establishment of this species to be aided by the establishment and spread of another species already in the Great Lakes?

Likely (A non-indigenous species to the Great Lakes that facilitates the development of this species—a major host, food item, pollinator—has already established and spread in the Great Lakes, AND available literature predicts this previous invader might promote the establishment of this species, AND/OR there have been cases reported of this species aiding the establishment of this species in other areas)	9

Somewhat likely (A non-indigenous species to the Great Lakes that facilitates the development of this species—a major host, food item, pollinator—has already established and spread in the Great Lakes)	6
Somewhat unlikely (A non-indigenous species to the Great Lakes that facilitates the development of this species—a major host, food item, pollinator—has already established in the Great Lakes BUT it is still confined to a small area of the Lakes and the likelihood of encounter with this species assessed is hard to predict)	3
Unlikely (A non-indigenous species to the Great Lakes that facilitates the development of this species has not been established in the Great Lakes)	0 √
Unknown	U
	0

- *A non-indigenous species in the Great Lakes has not been established that would facilitate the spread of the marbled crayfish.*

B14) How likely is establishment of this species to be prevented by the herbivory, predation, or parasitism of a natural enemy that is already present in the Great Lakes and may preferentially target this species?

Likely (The ability of the natural enemy to prevent the establishment of this species in introduced ranges or limiting populations of this species in native ranges is well documented in the literature AND this natural enemy is abundant and widespread in the Great Lakes)	-80% total points (at end)
Somewhat likely (The ability of the natural enemy to prevent the establishment of this species in introduced ranges or limiting populations of this species in native ranges is suggested in the literature OR this natural enemy has limited distribution in the Great Lakes)	-60% total points (at end)
Somewhat unlikely (There are few cases reported of such a natural enemy preventing the establishment of this species in introduced ranges or limiting populations of this species in native ranges OR this natural enemy has low abundance in the Great Lakes)	-10% total points (at end) √
Unlikely (Such a natural enemy is particularly rare or is not present in the Great Lakes)	0
Unknown	U
	-10%

- *Centrarchids and other predatory fish in the Great Lakes prey on crayfish, but are unlikely to be able to control a rapidly reproducing population (Hein et al. 2006; Aquiloni et al., 2010).*
- *Round goby (*Neogobius melanostomus*) is widely distributed in the Great Lakes and lab experiments suggest it is an efficient predator of* Procambarus virgnialis *(Franta et al. 2021; Roje et al. 2021). Round goby killed significantly more marbled crayfish than the other species used in experimental trials, suggesting a potential preference for marbled crayfish after satiation (Franta et al. 2021).* N. melanostomus *has also been shown to successfully compete with crayfish for shelter (Church et al. 2017). Predation and competition by round goby has potential to regulate marbled crayfish populations in the wild, but experts speculate that an eradication effect is less likely (Franta et al. 2021).*
- Procambarus virginalis *is speculated to respond inappropriately to a threat due to its origins as aquarium species, its response to a threat and boldness observed in the lab environment is different from other crayfish species (Linzmaier et al. 2018).*

PROPAGULE PRESSURE

B15) On average, how large and frequent are inoculations (introduction events) from the potential vectors identified in Section A for this species? (What is the total number of individuals introduced?)

Frequent, large inocula	9
Frequent, moderate inocula	6
Frequent, small inocula OR infrequent, large inocula	3
Infrequent, small or moderate inocula	0 √
Unknown	U
	0

- *Occasional aquarium releases by pet owners are the likely vector, and would not occur with great frequency or in high numbers (Souty-Grosset et al., 2006).*
- *Propagule pressure from active dispersal is also expected to be infrequent and small. The* Procambarus virginalis *population in Ontario would have to travel over land and through an urban environment to reach waters connected to the Great Lakes.*
- *Propagule pressure from* Procambarus virginalis *hitchhiking (i.e. bait bucket transfer or waterbird-mediated dispersal) is also expected to occur as infrequent, small incolua. Dispersal by waterbirds has not been documented for marbled crayfish and there are regulations restricting the use of marbled crayfish as bait in Ontario, limiting the potential for bait bucket transfer (Invasive Species Act, 2015, S.O. 2015, c. 22 - Bill 37).*

HISTORY OF INVASION AND SPREAD

B16) How extensively has this species established reproducing populations in areas outside its native range as a direct or indirect result of human activities?

Very extensively (many invasive populations of this species have been reported in areas widely distributed from the native range)	9 √
Extensively (some invasive populations of this species have been reported in areas widely distributed from the native range)	6
Somewhat extensively (few invasive populations of this species have been reported in areas widely distributed from the native range OR all invasive populations are in close proximity to each other)	3
Not extensively (no invasive populations of this species have been reported)	0
Unknown	U
	9

- *The spread of* Procambarus virginalis *has been very extensive and new populations have been identified in Europe and Canada over the last several years, new records of this species exist for Belgium (Scheers et al. 2021), Romania (Parvulescu et al. 2017), Israel (Carneiro et al. 2023), and Ontario (Hamr 2023). Additionally,* P. virginalis *populations in Madagascar are massive and can severely disrupt the ecosystem (Jones et al., 2009; Chucholl and Pfeiffer, 2010).*

B17) How rapidly has this species spread by natural means or by human activities once introduced to other locations?

Rapidly (This species has a history of rapid spread in introduced ranges)	9 √

Somewhat rapidly (This species has a history of moderately rapid spread in introduced ranges)	6
Somewhat slowly (This species has a history of moderately slow spread in its introduced ranges)	3
Slowly (This species has a history of slow to no spread in its introduced ranges)	0
Unknown	U
	9

- Procambarus virginalis *has a rapid rate of asexual reproduction, which has allowed it to colonize many waterways in Madagascar, and because crayfish can travel by land the risk is compounded. Human activity in Madagascar has also helped spread the marbled crayfish because it is used for food (Jones et al., 2009). With new populations emerging across Europe and elsewhere, Ercoli et al. (2019) state that marbled crayfish "is spreading alarmingly fast across European countries and beyond."*

B18) Are there any existing control measures in the Great Lakes set to prevent the establishment and/or spread of this species?

Yes, and they are likely to prevent establishment or spread of the species. (There are no reported cases of this species adapting or avoiding current measures. These measures are highly effective in preventing the establishment and spread of this species)	-90% total points (at end)
Yes, and they are moderately likely to prevent establishment or spread of the species. (There are few reported cases of this species adapting or avoiding current measures used to control its establishment and spread)	-50% total points (at end)
Yes, but they are unlikely to prevent establishment or spread of the species. (There are many reported cases of this species adapting or avoiding current measures used to control its establishment and spread)	-20% total points (at end) √
No control methods have been set to prevent the establishment and/or spread of this species	0
Unknown	U
	-20%

- *Methods used for other invasive crayfish may be applicable to this species (to limited effect), but once control measures are ceased the population is likely to return to its former level (Gherardi et al., 2011).*
- *There are currently no federal regulations in the United States for* Procambarus virginalis *but several states and provinces have passed regulations aimed to prevent its establishment and spread.*
    - *In Illinois,* Procambarus virginalis *" is not on the Illinois Aquatic Life Approved Species List and if it is not otherwise native to Illinois it is illegal to be imported or possessed alive without a permit" (515 ILCS 5/20-90).*
    - *In Michigan,* Procambarus virginalis *is prohibited and it is "unlawful to possess, introduce, import, sell or offer this species for sale as a live organism, except under certain circumstances" (Natural Resources Environmental Protection Act, Part 413 of Act 451).*
    - Procambarus virginalis *is regulated in Minnesota and "The transportation of live native and invasive crayfish from one waterbody to another within the state is prohibited, except by permit issued by the DNR. Live crayfish or crayfish eggs may not be imported without a permit issued by the DNR. Live crayfish may not be sold for live bait. Live crayfish can only be sold for aquarium use if the species cannot survive in Minnesota if released. Live crayfish taken from a waterbody can only be used as bait in that same waterbody" (Statute 84D.07).*
    - *In Ohio,* Procambarus virginalis *is prohibited and "it shall be unlawful for any person to possess, import or sell live individuals of this species" (Ohio Administrative Code 1501:31-19-01).*
    - Procambarus virginalis *is regulated in Pennsylvania, in this state "It is unlawful to sell, purchase, offer for sale, barter, possess, introduce, or import all species of live crayfish in Pennsylvania except when used: as bait on, in, or about the water from which taken; for testing and scientific*

purposes; for restaurant consumption when adequate measures have been taken to prevent their escape and they are accompanied by documentation stating the point of origin and the destination to which they are to be delivered" (58 Pa. Code § 63.46, 58 Pa. Code § 71.6).

- Procambarus virginalis *is regulated in Wisconsin, where all species of the family Cambaridae are prohibited and "one cannot transport, possess, transfer, or introduce it without a permit" (Chapter NR 40, Wis. Adm. Code).*
- Procambarus virginalis *"is prohibited in Ontario, making it illegal to import, possess, deposit, release, transport, breed/grow, buy, sell, lease or trade this species" (Invasive Species Act, 2015, S.O. 2015, c. 22 - Bill 37).*

- *However, it is important to note that there have been documented instances of people selling and distributing marbled crayfish despite current regulations (Milman 2022). Recent documentation of marbled crayfish in Ontario (Chandler 2023) suggests that release from aquaria remains a vector for introduction despite current regulations.*

Section B Scores				
Points	**Probability of Establishment**	**Total Points (pre-adjustment)**		114
>100	High	**Adjustments**		
		Critical species	A (114-0%)	114
51-99	Moderate	**Natural enemy**	B (114-10%)	103
		Control measures	C (103-20%)	82
0-50	Low	**Total Points (post-adjustment)**		**82**
		Probability of Establishment		**Moderate**
# of questions answered as "unable to determine"	**Confidence Level**			
0-1	High	**Total # of questions unknown**		**0**
2-5	Moderate			
6-9	Low	**Confidence Level**		**High**

Qualitative Statements for GLANSIS Fact Sheet:

***Procambarus virginalis* has a moderate probability of establishment if introduced to the Great Lakes (Confidence level: High).**

Comments:
In recent years, the spread of *Procambarus virginalis* has been prolific and new populations have been identified in Europe and Canada over the last several years, new records exist for Belgium (Scheers et al.

2021), Romania (Parvulescu et al. 2017), Israel (Carneiro et al. 2023), and Ontario (Hamr 2023). *P. virginalis* has a broad physiological tolerance and demonstrated the ability to survive a range of salinities, temperature, nutrient, and oxygen levels (Veselý et al. 2017; Vogt et al. 2019; Wiśniewski et al. 2020). Climate suitability analysis suggests that *P. virginalis* has a high climate match with the Great Lakes (USFWS 2023) and its persistence in Burlington, Ontario indicates it can tolerate the abiotic conditions present in the Great Lakes region (Hamr 2023). Predators present in the Great Lakes, such as the round goby *Neogobius melanostomus*, have the potential to regulate wild populations of *P. virginalis* but are not expected to prevent establishment (Franta et al. 2021). Ample regulations exist in the Great Lakes region to prevent the establishment and spread of *P. virginalis*, however documented instances of people selling and distributing marbled crayfish suggests current regulations may be insufficient to prevent establishment (Milman 2022). The high fecundity and parthenogenic reproduction of *P. virginalis* means a single individual can establish a new population without the need for a mate (Chucholl 2010).

SECTION C: POTENTIAL IMPACT ASSESSMENT RESULTS

Environmental: High
Socio-Economic: Moderate
Beneficial: High

Comments:

***Procambarus virginalis* has the potential for high environmental impact if introduced to the Great Lakes.**

As a crayfish of North American origin, *Procambarus virginalis* is a carrier of *Aphanomyces astaci*, the crayfish plague, is speculated to transmit the chytrid fungus Batrachochytrium dendrobatidis, and a carrier of rickettsiosis and coccidiosis (Souty-Grosset et al. 2006; Francesconi et al. 2021; Maciaszek et al. 2022). This species' fast growth rate and r-selected life history traits may allow it to outcompete native species (Jones et al. 2009; Chucholl and Pfeiffer 2010). *P. virginalis* can occupy multiple trophic levels and be a prey item for top level predators, it has the potential to significantly modify the food web of invaded systems (Vogt 2021). Classified as a tertiary burrower, *P. virginalis* may impact water quality, hydrology, and facilitate bank erosion, similar to the impacts observed for the closely related red swamp crayfish *P. clarkii (Rodríguez et al. 2005; Souty-Grosset et al. 2006; Kouba et al. 2016)*.

***Procambarus virginalis* has the potential for moderate socio-economic impact if introduced to the Great Lakes.**

The marbled crayfish is a burrowing species, and have been reported to damage irrigation systems and dams (Souty-Grosset et al. 2006), particularly in rice paddies, where they may also feed on and injure young plants (Jones et al. 2009; Kawai et al. 2009; Heimer 2010): similar effects may be possible in wild rice paddies around the Great Lakes. Fishermen in Madagascar also report severe negative impacts on their fishing grounds in the presence of marbled crayfish, but currently all evidence is anecdotal (Jones et al. 2009; Heimer 2010).

***Procambarus virginalis* has the potential for high beneficial impact if introduced to the Great Lakes.**

The marbled crayfish remains a popular pet species in Europe and North America (Chucholl 2010; Faulkes 2010), and is also used as fishing bait. *Procambarus virginalis* is a useful laboratory model organism for developmental physiology, epigenetics, and toxicology. Its large numbers of genetically identical offspring, rapid reproductive rate, and simple care requirements make it an ideal species for lab research (Vogt 2008; 2010). Recent publications document its increasing use as model organism for cancer research (Jirikowski et al. 2010; Rubach et al. 2011; Hossain et al. 2018).

POTENTIAL ENVIRONMENTAL IMPACT

C1) Does the species pose some hazard or threat to the health of native species (e.g., it magnifies toxin levels; is poisonous; is a pathogen, parasite, or a vector of either)?

Yes, and it has impacted threatened/endangered species, resulted in the reduction or extinction of one or more native populations, affects multiple species, or is a reportable disease	6 √
Yes, but negative consequences have been small (e.g., limited number of infected individuals, limited pathogen transmissibility, mild effects on populations and ecosystems)	1
Not significantly	0
Unknown	U

- *As a crayfish of North American origin, the marbled crayfish is a carrier of* Aphanomyces astaci*, the crayfish plague (Souty-Grosset et al., 2006; Francesconi et al. 2021; Dobrovic et al. 2022). The crayfish plague* A. astaci *is a reportable disease (USDA 2022).*
- *It is speculated that* Procambarus virginalis *can transmit the chytrid fungus* Batrachochytrium dendrobatidis*, which is pathogenic for amphibians (Maciaszek et al. 2022).*
- *Rickettsiosis and coccidiosis have both been found in marbled crayfish, and* Psorospermium *sp. is known to infect* P. fallax *in its indigenous range (Souty-Grosset et al., 2006).*

C2) Does it out-compete native species for available resources (e.g., habitat, food, nutrients, light)?

Yes, and it has resulted in significant adverse effects (e.g., impacted threatened/endangered species or caused critical reduction, extinction, behavioral changes including modified spawning behavior) on one or more native populations	6 √
Yes, and it has caused some noticeable stress to (e.g., decrease in growth, survival, fecundity) or decline of at least one native population	1
Not significantly	0
Unknown	U

- *Marbled crayfish pose a threat to indigenous crayfish species in Madagascar and Europe due to competition for food and space and crayfish plague transmission (Jones et al. 2009; Kawai et al. 2009; Chucholl and Pfeiffer 2010).*
- *Marbled crayfish reproduce by parthenogenesis, which makes the risk of release resulting in a reproducing population considerably greater than for sexually reproducing species: a single individual is sufficient to create a new population (Kawai et al 2009).*
- *Current data suggests that the marbled crayfish is a fast-growing species that exhibits r-selected life history traits like early maturation, an extended breeding period, and high fecundity, giving it a competitive advantage over other species (Seitz et al. 2005; Jones et al. 2009; Chucholl and Pfeiffer 2010).*
- *Results of a dynamic energy budget model suggest that marbled crayfish is a superior competitor when considering metabolism and ontogeny. Models show that marbled crayfish has a clear potential to be a better competitor than* Astacus astacus*,* Austropotamobius torrentium*, and* Pacifastacus leniusculus *(Marn et al. 2022).*
- *A lab experiment examining agonistic interactions between marbled crayfish* (Procambarus virginalis) *and red swamp crayfish* (Procambarus clarkii) *suggests that marbled crayfish has a competitive advantage. Marbled crayfish won significantly more fights and more frequently established dominance in both premature and mature experimental trials (Hossain et al. 2019).*
- *Marbled crayfish has shown competitive dominance over calico crayfish (*Faxonius immunis*) in experimental lab studies. When no shelter was available, marbled crayfish won significantly more fights.*

With available shelter, marbled crayfish dominance was 100% for female and 54.5% for male calico crayfish (Hossain et al. 2020).

C3) Does it alter predator-prey relationships?

Yes, and it has resulted in significant adverse effects (e.g., impacted threatened/endangered species, caused significant reduction or extinction of one or more native populations, creation of a dead end or any other significant alteration in the food web)	6
Yes, and it has resulted in some noticeable stress to (e.g., decrease in growth, survival, fecundity) or decline of at least one native population AND/OR Yes, and it has resulted in some alteration of the food web structure or processes, the effects of which have not been widespread or severe	1 √
Not significantly	0
Unknown	U

- Procambarus virginalis *can occupy multiple trophic levels and be a prey item for top level predators, it has the potential to significantly modify the food web of invaded systems (Vogt 2021).*
- *Anecdotal evidence suggests that the introduction of* P. virginalis *in a nature reserve in Poland led to a noticeable decline in frog species, with reports of decreased croaking activity throughout the forest (Maciaszek et al. 2022).*
- *Results of an experimental study suggest that* P. virginalis *alters the food web by increasing pressure on macrophytes, macrophyte-dependent organisms, and influencing the detritus decomposition (Linzmaier et al. 2020).*

C4) Has it affected any native populations genetically (e.g., through hybridization, selective pressure, introgression)?

Yes, and it has caused a loss or alteration of genes that may be irreversible or has led to the decline of one or more native species (or added pressure to threatened/endangered species)	6
Yes, some genetic effects have been observed, but consequences have been limited to the individual level	1
Not significantly	0 √
Unknown	U

- *The marbled crayfish is triploid, and attempted crossbreeding experiments with* P. fallax *and* P. alleni *indicate that they are incapable of hybridization: all offspring produced even after mating with males are identical female marbled crayfish with no genetic contribution from the males (Vogt et al. 2015).*

C5) Does it negatively affect water quality (e.g., increased turbidity or clarity, altered nutrient, oxygen, or other chemical levels/cycles)?

Yes, and it has had a widespread, long-term, or severe negative effect on water quality AND/OR Yes, and it has resulted in significant negative consequences for at least one native species	6
Yes, it has affected water quality to some extent, but the alterations and resulting adverse effects have been limited or inconsistent (as compared with above statement)	1 √
Not significantly	0
Unknown	U

- *This species is a tertiary burrower (Kouba et al. 2016), and while only anecdotal evidence exists of how the marbled crayfish in particular may alter water quality, the related species* Procambarus clarkii *has had*

major impacts on native ecosystems and degraded shallow wetlands on the Iberian Peninsular within a few years. In Lake Chozas, the invasion by P. clarkii *led to a switch from a clear water state to a turbid one, followed by a severe biodiversity reduction (Rodríguez et al. 2005).*

C6) Does it alter physical components of the ecosystem in some way (e.g., facilitated erosion/siltation, altered hydrology, altered macrophyte/phytoplankton communities, physical or chemical changes to substrate)?

Yes, and it has had a widespread, long term, or severe negative effect on the physical ecosystem AND/OR Yes, and it has resulted in significant negative consequences for at least one native species	6
Yes, it has affected the physical ecosystem to some extent, but the alterations and resulting adverse effects have been mild	1 √
Not significantly	0
Unknown	U

- *Marbled crayfish are polytrophic omnivores and were found at very high densities in Madagascar (Jones et al. 2009). Given that, they may have a very significant impact on ecosystem functioning and integrity, although specific and quantified information is currently lacking.*
- Procambarus virginalis *is classified as a tertiary burrower, this species may impact hydrology and facilitate bank erosion (Kouba et al. 2016).*

Environmental Impact Total	**15**
Total Unknowns (U)	**0**

Scoring		
Score	# U's	**Impact**
>5	Any	**High**
2-5	Any	**Moderate**
0	0-1	**Low**
1	0	
0	≥2	**Unknown**
1	≥1	

POTENTIAL SOCIO-ECONOMIC IMPACT

C7) Does the species pose some hazard or threat to human health (e.g., it magnifies toxin levels, is poisonous, a virus, bacteria, parasite, or a vector of one)?

Yes, significant effects on human health have already been observed	6
Yes, but negative consequences have not been widespread, long lasting, or severe	1
Not significantly	0 √
Unknown	U

- *At the time of this review, there is no evidence to suggest that* Procambarus virginalis *poses a hazard or threat to human health.*

C8) Does it cause damage to infrastructure (e.g., water intakes, pipes, or any other industrial or recreational infrastructure)?

Yes, it is known to cause significant damage	6
Yes, but the costs have been small and are largely reparable or preventable	1 √
Not significantly	0
Unknown	U

- Procambarus virginalis *digs burrows which may cause damage to irrigation systems and dams (Souty-Grosset et al. 2006; Kouba et al. 2016).*

C9) Does it negatively affect water quality (i.e. in terms of being less suitable for human use)?

Yes, it has significantly affected water quality, and is costly or difficult to reverse	6
Yes, but the effects are negligible and/or easily reversed	1
Not significantly	0 √
Unknown	U

- Procambarus virginalis *burrowing may cause increased turbidity, but at the time of this review there is no evidence to suggest this species negatively impacts water quality in terms of suitability for human use.*

C10) Does it negatively affect any markets or economic sectors (e.g., commercial fisheries, aquaculture, agriculture)?

Yes, it has caused significant damage to one or more markets or economic sectors	6
Some damage to markets or sectors has been observed, but negative consequences have been small	1 √
Not significantly	0
Unknown	U

- *Anecdotal observations from Madagascar suggest a significant impact on fish populations. Local fishermen reported that marbled crayfish have destroyed fishing in their area (Jones et al., 2009; Heimer, 2010).*
- *There is serious concern in Madagascar that the invasion of marbled crayfish will negatively impact rice culture by feeding on and destroying young plants (Jones et al., 2009; Kawai et al., 2009; Heimer, 2010), which suggests that they may also have the potential to negatively impact wild rice cultivation in the Great Lakes if introduced to paddies.*
- *A socio-economic survey in Madagascar reports negative impacts on rice farming from marbled crayfish burrowing. Negative impacts were reported across all sites and linked to marbled crayfish burrows destroying the mud walls around the rice fields (Andriantsoa et al. 2020).*

C11) Does it inhibit recreational activities and/or associated tourism (e.g., through frequent water closures, equipment damage, decline of recreational species)?

Yes, it has caused widespread, frequent, or otherwise expensive inhibition of recreation and tourism	6
Yes, but negative consequences have been small	1
Not significantly	0 √
Unknown	U

- Procambarus virginalis *burrowing may increase turbidity but at the time of this review there is no evidence to suggest that this species has negatively impacted recreational activities or associated tourism.*

C12) Does it diminish the perceived aesthetic or natural value of the areas it inhabits?

Yes, the species has received significant attention from the media/public, significantly diminished the natural or cultural character of the area, or significantly reduced the area's value for future generations	6
Yes, but negative consequences have been small	1
Not significantly	0 √
Unknown	U

- *Significant media coverage in February 2018 around the globe has led to increased public awareness of marbled crayfish, but no significant impacts on natural or aesthetic value have been observed.*

Socio-Economic Impact Total	**2**
Total Unknowns (U)	**0**

Scoring		
Score	# U's	**Impact**
>5	Any	**High**
2-5	Any	**Moderate**
0	0-1	**Low**
1	0	
0	≥2	**Unknown**
1	≥1	

POTENTIAL BENEFICIAL EFFECT

C13) Does it act as a biological control agent for aquatic weeds or other harmful nonindigenous organisms?

Yes, it has succeeded significantly as a control agent	6
Yes, it has had some success as a control agent, but may be inconsistent or lack a desired level of effectiveness	1
Not significantly	0 √
Unknown	U

- *Field studies in Madagascar suggest that marbled crayfish has the potential to be used as a biocontrol agent against the freshwater snail* Biomphalaria pfeifferi*, which serves as an intermediate host of Schistosoma flatworms (Andriantsoa et al. 2019). However, at the time of this review there is no evidence to suggest that marbled crayfish has been intentionally utilized as a biocontrol agent.*

C14) Is it commercially valuable (e.g., for fisheries, aquaculture, agriculture, bait, ornamental trade)?

Yes, it is economically important to at least one of these industries	6
Yes, but its economic contribution is small	1 √
Not significantly	0
Unknown	U

- *In Madagascar, marbled crayfish are sold in markets for human consumption (Jones et al., 2009, Kawai et al., 2009; Heimer, 2010). However, Heimer (2010) and Jones et al. (2009) both indicate that marbled crayfish are of low economic value, and their popularity as a food source is limited by their small size and the subsequent amount of work it takes to clean them for eating.*
- *This species is a popular and inexpensive aquarium pet: they are available for purchase from hobbyists in the United States for about $10-15 per specimen (Aquatic Arts, 2018).*
- *A socio-economic survey in Madagascar found that marbled crayfish has been accepted as a food source. Many people report catching and selling crayfish, or using them as a cheap source of household protein or animal feed. Other responses indicate a perceived positive impact on rice farming because marbled crayfish burrowing loosens and aerates the soil (Andriantsoa et al. 2020).*
- *In Ukraine, invasive populations of* P. virginalis *were quickly utilized as a food resource and harvested in large numbers to sell at local markets (Son et al. 2020).*

C15) Is it recreationally valuable (e.g., for sport or leisurely fishing, as a pet, or for any other personal activity)?

Yes, it is commonly employed recreationally and has some perceived value for local communities and/or tourism	6
It is sometimes employed recreationally, but adds little value to local communities or tourism	1 √
Not significantly	0
Unknown	U

- Procambarus virginalis *remains a popular pet species in Europe and North America (Chucholl 2010; Faulkes 2010; Blaha et al. 2022; Lipták et al. 2023).*
- P. virginalis *is a popular and inexpensive aquarium pet, they are available for purchase from hobbyists in the United States for about $5-25 per specimen (The Shrimp Farm 2023).*

C16) Does the species have some medicinal or research value (i.e. outside of research geared towards its control)?

Yes, it has significant medicinal or research value	6 √
It has some medicinal or research value, but is not of high priority OR It is potentially important to medicine or research and is currently being or scheduled to be studied	1
Not significantly	0
Unknown	U

- *The marbled crayfish is a useful laboratory model organism for developmental physiology, epigenetics and toxicology. Its large numbers of genetically identical offspring, fast reproductive rate, and simple care requirements make it an ideal species for lab research (Vogt 2008; 2010). Recent publications document its increasing use as a model organism (e.g. Jirikowski et al. 2010; Rubach et al. 2011; Hossain et al. 2018).*

C17) Does the species remove toxins or pollutants from the water or otherwise increase water quality?

Yes, it reduces water treatment costs or has a significant positive impact for the health of humans and/or native species	6
Yes, but positive impact for humans or native species is considered negligible	1
Not significantly	0 √
Unknown	U

- *To the contrary, the burrowing behavior of* Procambarus virginalis *may increase turbidity (Kouba et al. 2016).*

C18) Does the species have a positive ecological impact outside of biological control (e.g., increases the growth or reproduction rates of other species, fills an important gap in the food web, supports the survival of a species that is threatened, endangered species, or commercially valuable)?

Yes, it significantly contributes to the ecosystem in one or more of these ways	6
Yes, it provides some positive contribution to the ecosystem, but is not vital	1
Not significantly	0 √
Unknown	U

- *Any benefits* Procambarus virginalis *may provide as a food source for large fish or other organisms are likely canceled out by its competition with native crayfish and other disruptive behaviors.*

Beneficial Effect Total	**8**
Total Unknowns (U)	**0**

Scoring		
Score	# U's	**Impact**
>5	Any	**High**
2-5	Any	**Moderate**
0	0-1	**Low**

1	0	
0	≥2	**Unknown**
1	≥1	

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